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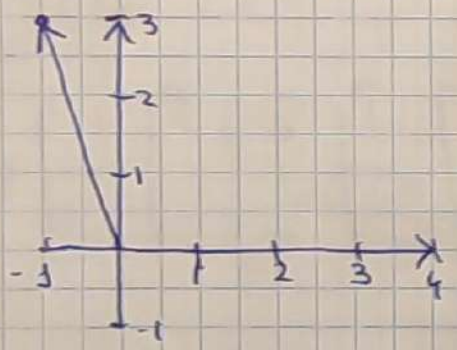
24B031672

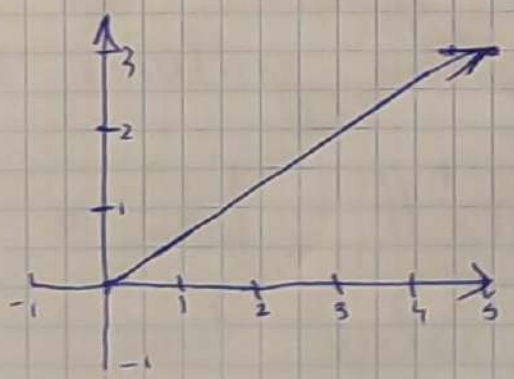
1) State the type and dimensionality of the following vectors (e.g "four-dimensional column vector"). For 2D vectors, additionally draw the vector starting from the origin.

a. $\begin{bmatrix} 1 \\ 2 \\ 3 \\ 1 \end{bmatrix}$ - four-dimensional column vector

b. $[-1, \ 3]$ - two-dimensional row vector



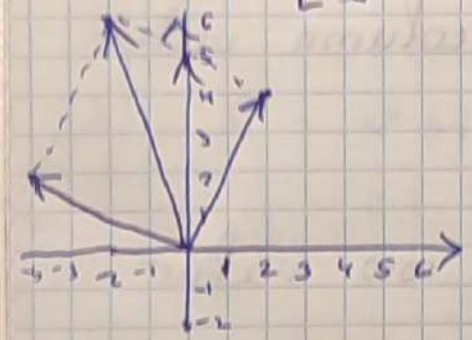
c. $\begin{bmatrix} 5 \\ 3 \end{bmatrix}$ - two dimensional column vector



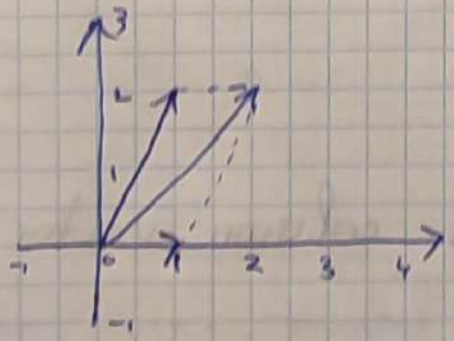
2) Solve the following operations. For 2D vectors, draw both vectors in standart position, and the vector sum (also in standart position)

a. $[-1 \quad 2 \quad 5 \quad 4] + [-3 \quad 4 \quad 3 \quad 6] = [-4 \quad 6 \quad 8 \quad 10]$

b. $\begin{bmatrix} 2 \\ 4 \end{bmatrix} + \begin{bmatrix} -4 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ 6 \end{bmatrix}$



c. $\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$



3) Compute dot product between the following pairs of vectors

a. $\begin{bmatrix} -4 \\ -2 \end{bmatrix}^T \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} -4, & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = -4 + (-6) = -10$

b. $\begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}^T \begin{bmatrix} -2 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0, & 1 & 2 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} = 0 - 1 + 0 = -1$

4) Compute the outer product between the following pairs of vectors

a. $\begin{bmatrix} 1 \\ 2 \end{bmatrix}^T \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$

b. $\begin{bmatrix} -1 \\ 3 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}^T = \begin{bmatrix} -1 \\ 3 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 3 \\ 0 & 3 & 9 \\ 0 & 0 & 0 \end{bmatrix}$

5) Compute a unit vector in the same direction as the following vectors. Compute the magnitude of resulting vector to make sure it equal 1

a. $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$ $\quad \left\| \begin{matrix} 1 \\ 3 \end{matrix} \right\| = \sqrt{1^2 + 3^2} = \sqrt{10}$

unit vector $= \frac{1}{\sqrt{10}} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{10}} \\ \frac{3}{\sqrt{10}} \end{bmatrix}$

$$\left\| \begin{matrix} \frac{1}{\sqrt{10}} \\ \frac{3}{\sqrt{10}} \end{matrix} \right\| = \sqrt{\frac{1}{10} + \frac{9}{10}} = 1$$

b. $\begin{bmatrix} 6 \\ -8 \end{bmatrix}$ $\quad \left\| \begin{matrix} 6 \\ -8 \end{matrix} \right\| = \sqrt{36 + 64} = 10$

unit vector $= \frac{1}{10} \begin{bmatrix} 6 \\ -8 \end{bmatrix} = \begin{bmatrix} \frac{3}{5} \\ -\frac{4}{5} \end{bmatrix}$

$$\left\| \begin{matrix} \frac{3}{5} \\ -\frac{4}{5} \end{matrix} \right\| = \sqrt{\frac{9}{25} + \frac{16}{25}} = 1$$